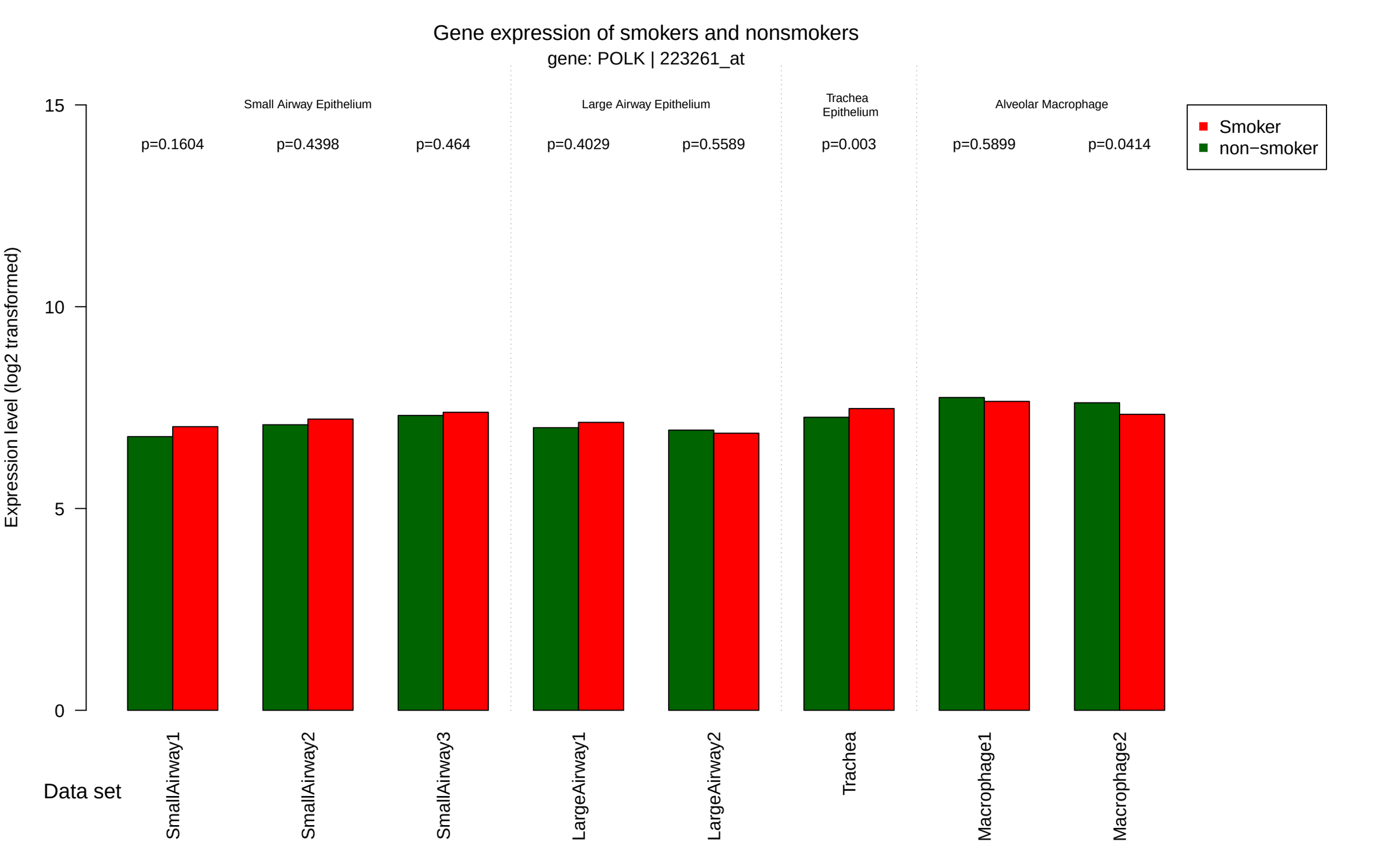
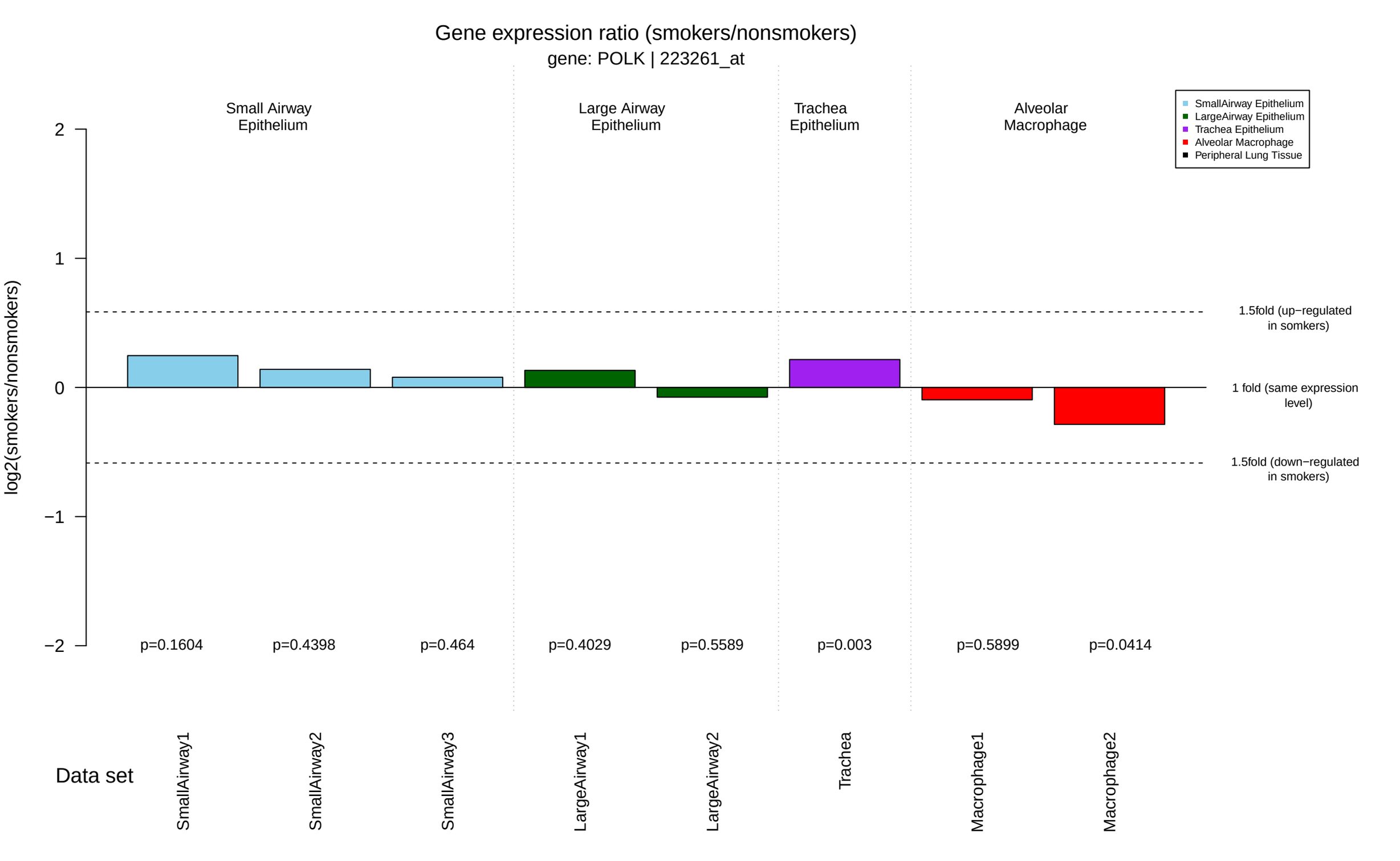
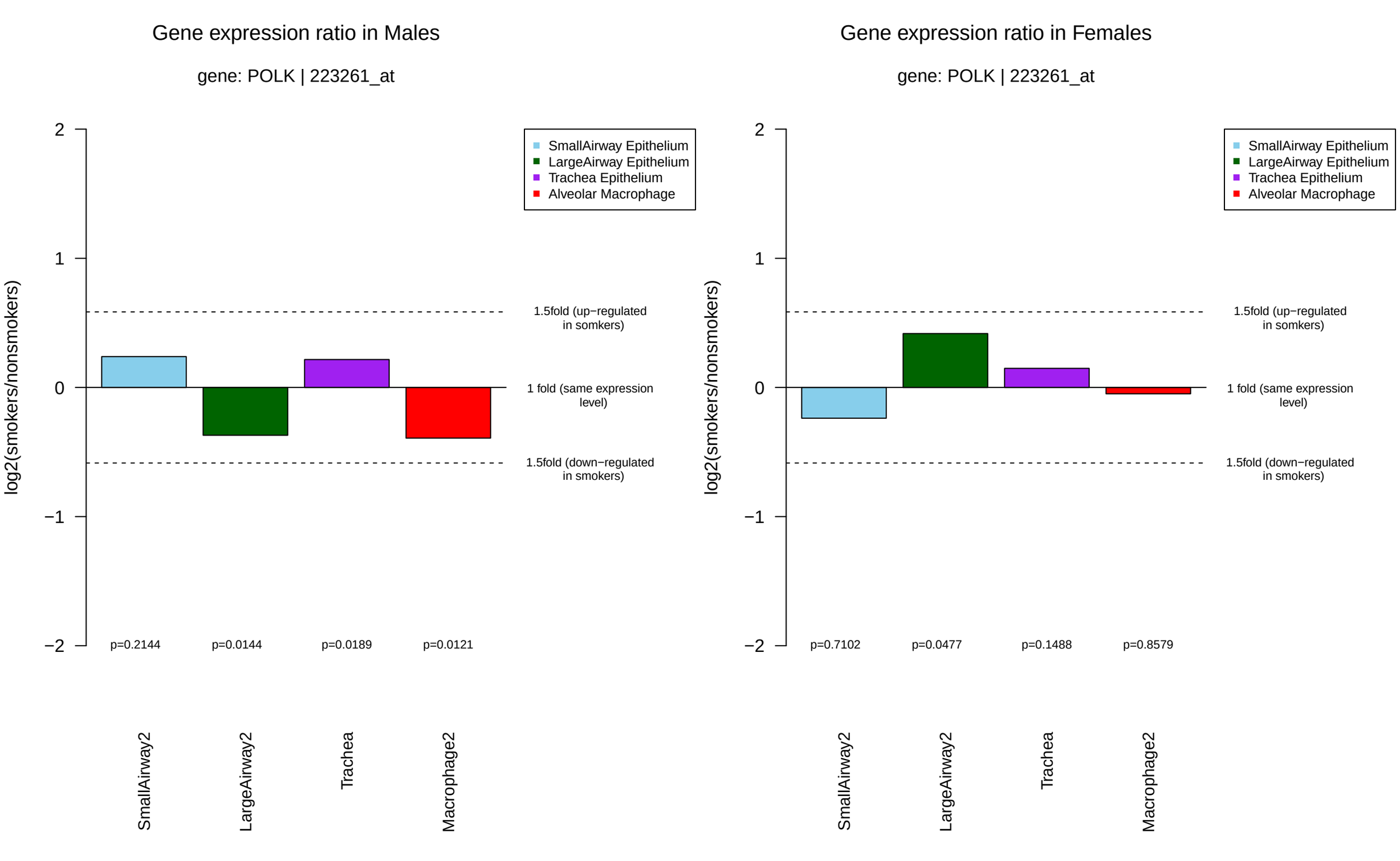
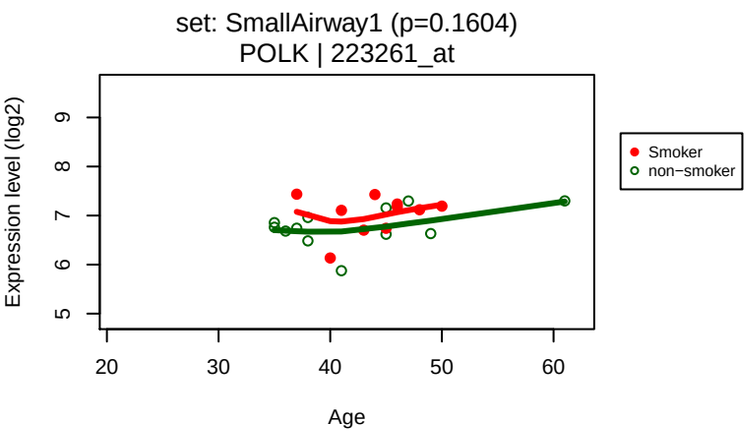
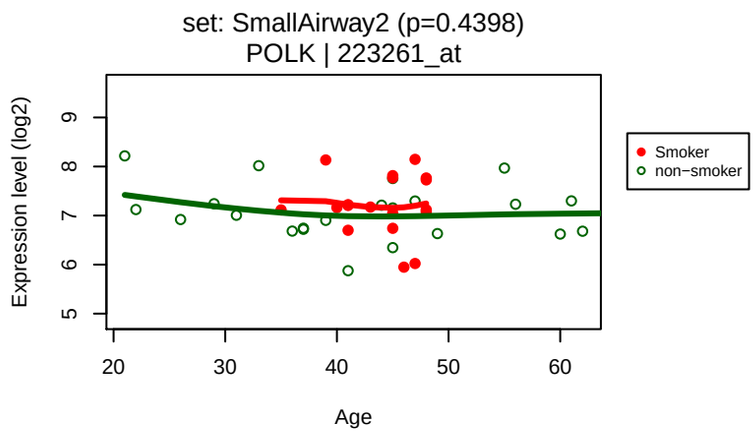
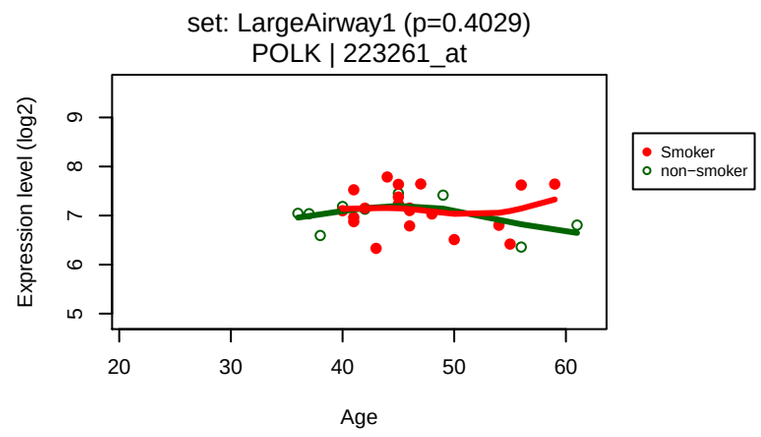

Gene expression of smokers and nonsmokers
gene: POLK | 223261_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1604
p=0.4398
p=0.464
p=0.4029
p=0.5589
p=0.003
p=0.5899
p=0.0414
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

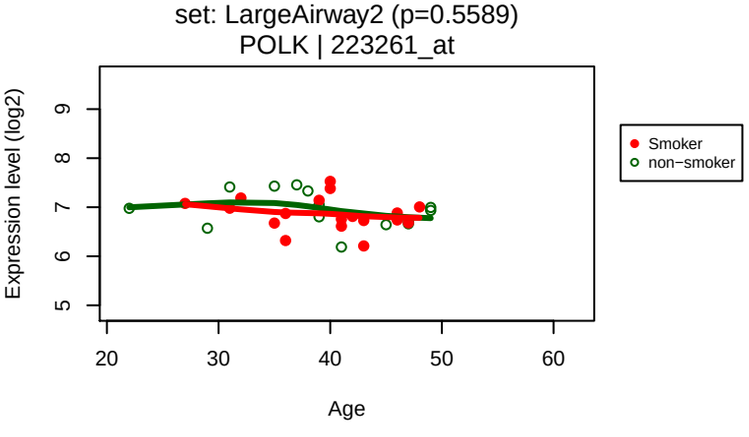

Gene expression ratio (smokers/nonsmokers)
gene: POLK | 223261_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1604
p=0.4398
p=0.464
p=0.4029
p=0.5589
p=0.003
p=0.5899
p=0.0414
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

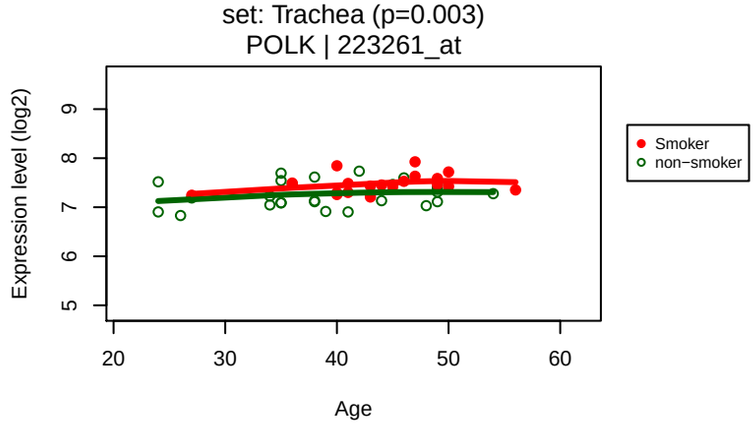

Gene expression ratio in Males
gene: POLK | 223261_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2144
p=0.0144
p=0.0189
p=0.0121
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLK | 223261_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7102
p=0.0477
p=0.1488
p=0.8579
SmallAirway2
LargeAirway2
Trachea
Macrophage2

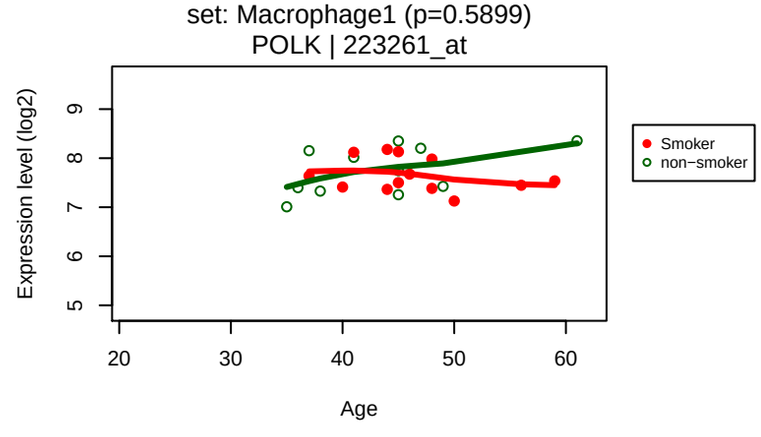

set: SmallAirway1 (p=0.1604)
POLK | 223261_at
Expression level (log2)
Age
Smoker
non−smoker

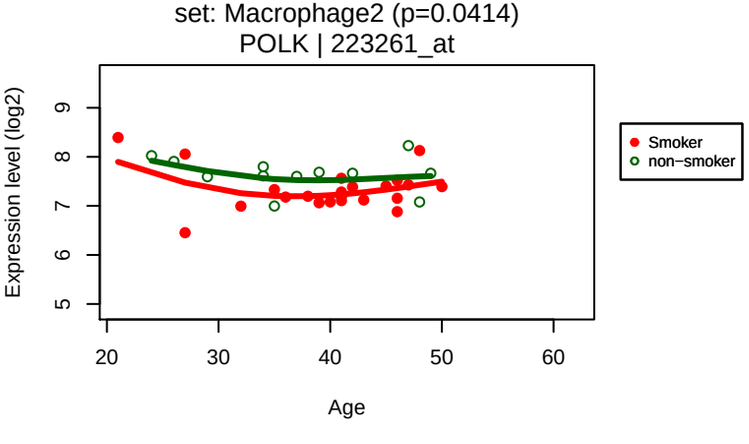

set: SmallAirway2 (p=0.4398)
POLK | 223261_at
Expression level (log2)
Age
Smoker
non−smoker

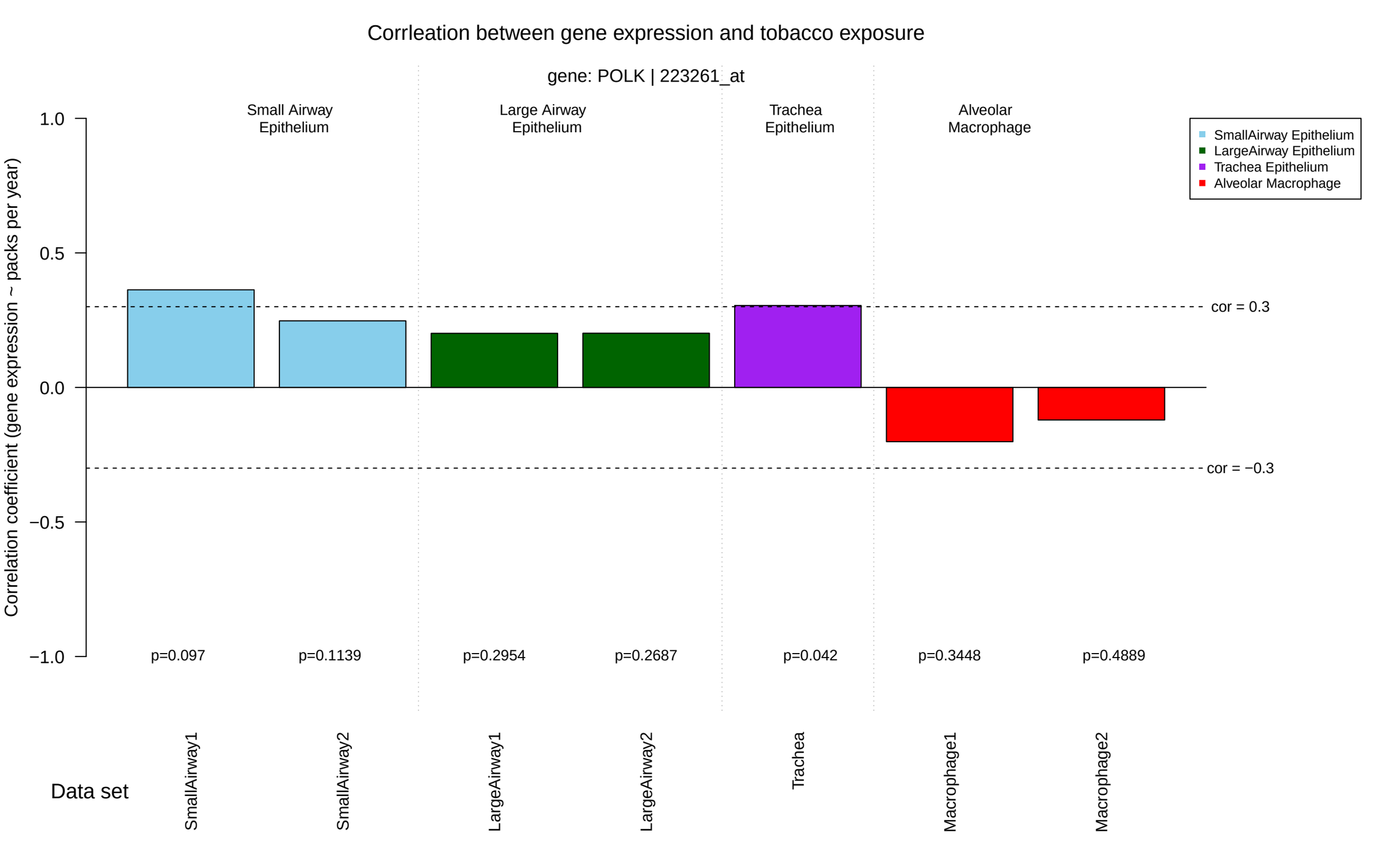

set: LargeAirway1 (p=0.4029)
POLK | 223261_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5589)
POLK | 223261_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.003)
POLK | 223261_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5899)
POLK | 223261_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0414)
POLK | 223261_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLK | 223261_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.097
p=0.1139
p=0.2954
p=0.2687
p=0.042
p=0.3448
p=0.4889
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2